

Choice Without a Central Chooser

A Falsifiable Multiscale Model of Neural Route Selection

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

July 2026

Abstract

Neural tissue contains no single executive that inspects every available action, yet an organism can maintain alternatives, commit to one action, monitor its consequences, and alter later behavior. This paper proposes a multiscale model in which learned synaptic and dendritic states make cellular transitions criterion-sensitive; excitation, inhibition, and disinhibition change time-varying effective connectivity; distributed cortical and subcortical dynamics transform competing action-related states; and action-conditioned sensory feedback updates the criteria governing later transitions. Four biological scales are kept distinct: cellular transition, circuit route selection, population commitment, and organism action. The account credits prior work on recurrent path diversion, criterial causation, evidence accumulation, affordance competition, context-dependent dynamics, basal-ganglia and superior-colliculus selection, thalamic switching, cerebellar evidence processing, dendritic events, corollary discharge, and reafference. Its candidate contribution is the joined, testable ladder from learned criteria to trial-specific route, commitment, controlled action, consequence, and later updating.

This final-preprint successor closes the primary-page locator for the paper's retained Tse atoms, states the influence, extension, and philosophical departure without absorbing criterial causation into SAN, and freezes an analysis plan for a prospective route screen in the International Brain Laboratory Brain Wide Map release. The screen is observational: it can reject or prioritize route candidates but cannot establish inhibitory causation, organism-level agency, or the complete multiscale ladder. The draft preserves the preparation-bounded subcortical comparisons, internal adversarial ledger, Buzsaki Figure 11.5 boundary, 2017 protocol-search genealogy, and deterministic synthetic fixture. Across 30 replicates per scenario, a prospective true-route variable improved held-out log loss by 0.0352 and passed the declared gate in 30/30 runs. Route-null and global-gain fixtures passed in 0/30. A post-outcome leakage variable produced an apparent gain of 0.5525 but was rejected in 30/30 because its provenance was invalid. An added timing feature was removed in 30/30 timing-null runs. These are software checks, not biological results. Receiver-relative phase, duration, waveform, magnitude, and spatial relations remain optional and must outperform anatomy, rate, power, movement, arousal, common input, and capacity-matched recurrent baselines. The proposal does not establish phenomenal consciousness, metaphysical freedom, moral responsibility, or one universal neural circuit. It also introduces a narrower bridge from coincident, receiver-compatible population activity to signed effective drive, a departure from tonic expectation, motor realization, and returned consequence. The terms *electrical valence* and *building phase-wave differential* are introduced only after that operation is stated in ordinary physiological variables. They are removable hypotheses, not synonyms for psychological valence, spectral power, bulk electric charge, or muscular energy.

inhibition; dendritic computation; population dynamics; agency; reafference

Keywords: neural choice; action selection; effective connectivity;

Scope and evidence boundary

selfawarenetworks.com

1. The Explanatory Problem

A reader can continue reading, look away, speak, stand, or remain still. Only one of those possibilities becomes the next overt action, but no neuron is shown a complete menu. A mechanistic account therefore needs to explain four things without installing a smaller decision-maker inside the brain:

1. how more than one action remains physically and computationally available;
2. how present evidence, learned criteria, body state, and predicted consequences alter their relative support;
3. how distributed activity crosses a commitment boundary and controls a bounded action; and
4. how the action's consequences change the next state of the system.

The final button press is not the whole choice. Neither is one threshold crossing, one inhibitory event, or one decoded action label. The object of study is a temporally ordered causal sequence linking multiple organizational scales.

2. Four Biological Scales

2.1 Cellular transition

At the cellular scale, “choice” is permissible only as shorthand for a criterion-sensitive transition among measured states: no event, a dendritic event, a spike, a burst, altered release, or a plasticity event. Morphology, synaptic efficacy, receptor and channel state, membrane potential, neuromodulation, metabolism, and input timing can change the transition. The cell is not thereby assigned personhood, deliberation, or moral agency.

2.2 Circuit route selection

Gross anatomy defines possible interactions. Current synaptic efficacy, release probability, excitability, inhibition, disinhibition, timing, and input determine which interactions exert causal influence on a trial. The paper calls this a time-bounded effective route. It is not a second anatomical tract and need not be a unique serial chain.

2.3 Population commitment

Distributed activity can contain partially coexisting action-related states. Depending on the preparation, these may be described as cell assemblies, population subspaces, recurrent trajectories, or decision variables. Commitment is a prospective transition toward stable control of one action, not retrospective classification alone.

2.4 Organism action

An organism-level choice occurs when a bounded sensorimotor system moves from declared alternatives to controlled behavior under its current evidence, policy, body state, and context. The resulting visual, proprioceptive, tactile, vestibular, and interoceptive consequences return to the system and can alter its next policy or criteria.

Philosophical and legal responsibility are not a fifth neural scale. They are interpretations of capacities and social rules that require additional arguments beyond this mechanism.

3. Prior Components and Attribution

Buzsaki describes hippocampal activity trajectories shaped by recurrent weights and momentary local inhibition, including diversion at a bifurcation [1]. This is an important prior description of state-dependent path selection. It is not by itself a general theory of organism-level choice, action ownership, or conscious deliberation.

Tse’s criterial-causation framework argues that physically implemented information can modify the criteria governing later neural events [2]. That criteria-update atom is prior to the present synthesis. The current paper does not claim to originate criterial causation.

3.1 Tse adoption, extension, and divergence

The official MIT Press / Oxford record supplies a firmer boundary than a later paraphrase. Chapter 3, especially Sections 3.9-3.10 (pp. 25-26), presents a three-stage criterial sequence: criteria are set or reset, variable input later meets or fails those criteria, and a physical event follows. Tse explicitly distinguishes the present physical realization from changes that prepare future decoders or criteria in the same or other neurons. Chapter 4, especially Sections 4.1-4.4 (pp. 31-34), develops neuronal firing as the satisfaction of physical and informational criteria, including coincidence-sensitive examples [2]. The publisher abstract locates this program across NMDA receptors, synapses, dendrites, neurons, and circuits and links it to rapid synaptic plasticity, downward mental causation, and a strong form of free will.

Relation	What this paper retains	What this paper changes or declines
adoption	physically implemented criteria can condition later neural transitions	no claim that the present paper originated criterial causation
cellular extension	a cellular event is modeled as a criterion-sensitive state transition	the cell is not a person, deliberator, or complete chooser
circuit extension	criteria are linked to time-varying downstream accessibility and population commitment	the link is a hypothesis requiring mediation and perturbation, not a consequence of Tse’s framework alone
organism extension	consequences may alter later criteria and policies	reward, error, reafference, and explicit rule learning remain rival update routes
philosophical divergence	information has causal relevance only through physical state and update rules	this paper does not require indeterminism, downward causation as a separate force, or proof of metaphysical free will

Micah’s owner-dated 2014 note paraphrases this as a neuron being unable to change the choice it is presently realizing while its activity can alter later criteria. The primary text supports the temporal distinction but is more precise: a current physical realization does not change itself; it can prepare or alter future criteria in the same or other neurons. The 2014 note is therefore retained as Micah’s interpretation, not quoted as Tse’s sentence. The 2018 public article then adds Micah’s bounded claim that each cell may make a “choice in a sense” through thresholded criteria, while explicitly placing more complex choice in circuits.

The intellectual relationship is adoption followed by extension and departure. This paper credits Tse with criterial causation, coincidence-sensitive firing, and information-dependent changes to later criteria. Micah’s subsequent synthesis joins those components to deterministic organism-level responsibility, coincidence as a neural information event, inhibitory branch switching, dendritic and apical action configurations, oscillatory phase routing, perception-consideration-action feedback, and NAPOT rendering [18]. Those joined operations remain hypotheses where they exceed the support of the cited experiments. The philosophical departure is also narrow: randomness or quantum unpredictability is not treated as sufficient for

agency, and the model does not require indeterminacy as its foundation. The broader free-will conclusions of either author remain outside this paper’s required premises.

Decision neuroscience supplies additional indispensable components. Multiple reach options can be represented before final selection [3], affordance competition provides a distributed action-selection framework [4], recurrent dynamics and accumulators explain evidence integration and commitment [5,6], and context-dependent population trajectories can implement flexible selection [7]. Readiness-potential work warns against identifying averaged preparation with one fixed decision time [8]. Sense of agency must also be separated from causal control [9].

The candidate contribution is therefore not inhibition, thresholds, competition, prediction, or feedback in isolation. It is the declared scale ladder:

```

learned criteria
-> criterion-sensitive cellular transition
-> time-bounded effective route
-> competing population dynamics
-> action commitment
-> motor command and predicted consequence
-> reafferent consequence
-> later criterion or policy update

```

Each arrow is a hypothesis with its own observation, intervention, and failure condition. The whole ladder can be narrowed without denying its established components.

A direct perturbational anchor strengthens one arrow while preserving that rule. In mouse hippocampal CA1 preparations, Lovett-Barron et al. used optical activation and cell-type-specific silencing to show that dendrite-targeting inhibition regulates principal-cell input-output transformation by gating the dendritic electrogenesis that drives burst spiking [20]. This supports an identified inhibitory-control mechanism between synaptic input and cellular output. It does not establish organism-level choice, a unique neural route, or the complete criteria-to-action ladder.

4. From Structure to Effective Route

Let the structural graph be $G = (V, E)$. For a connection from unit i to unit j , define a trial- and time-dependent effective coupling:

$$g_{ij}^{\text{eff}}(t) = A_{ij}w_{ij}(t)r_{ij}(t)\gamma_{ij}(t, \mathbf{c}_t). \quad (1)$$

Here A_{ij} denotes structural availability, w_{ij} synaptic efficacy, r_{ij} a preparation-specific release or excitability term, and γ_{ij} compatibility with timing and context $\mathbf{c}_t$. This is bookkeeping, not a claim that these factors are independent or simultaneously observable in every preparation. Candidate measurements include structural tracing, paired or population perturbation, synaptic efficacy, release indicators, membrane or dendritic voltage, identified cell-class activity, and time-resolved effective-connectivity estimates.

An inhibitory contribution must retain cell class, target, and timing:

$$I_j(t) = \sum_k h_{kj}^{(q_k, \ell_k)} z_k(t - \tau_{kj}), \quad (2)$$

where q_k is an identified interneuron class, ℓ_k is the target compartment or population, and τ_{kj} is the delay. This expression blocks a misleading railway interpretation. An interneuron does not redirect a spike into another axon. It can alter gain, timing, excitability, competition, stability, and downstream accessibility. Disinhibitory circuits and choice-selective inhibitory motifs establish that such effects are testable in specific preparations [12-17]; they do not establish one universal switch.

The CA1 result in [20] also sharpens what a route claim must mean. The measured effect was compartment- and cell-class-dependent control of dendritic electrogenesis and burst output, not physical redirection of an action potential. In the present model, a route is therefore an intervention-sensitive change in which downstream states remain dynamically accessible. Evidence for that route requires target engagement, a predicted downstream redistribution, and exclusion of global gain, arousal, sensory loss, and motor impairment.

5. From Effective Route to Commitment

Let $\mathbf{h}_t$ summarize current sensory evidence, recurrent expectation, body state, and action-history or reafferent context. Population state evolves as

$$\mathbf{x}_{t+\Delta} = F(\mathbf{x}_t, \mathbf{u}_t, \mathbf{c}_t, \mathbf{h}_t, G_t^{\text{eff}}) + \epsilon_t. \quad (3)$$

The baseline F may be a capacity-matched recurrent model. The proposal earns content only when an identified route variable improves prediction or causal control beyond that baseline.

For action a , let $q_a(t)$ be a decoded and, where possible, causally tested support variable. Commitment is represented as a transition:

$$\kappa_t : \{q_a(t) : a \in \mathcal{A}_t\} \longrightarrow (a_t^*, t_c, \rho_t), \quad (4)$$

where $\mathcal{A}_t$ is the task-declared action set, t_c is a commitment interval, and ρ_t measures stability, confidence, or control under the task. Equation (4) is not a global argmax and does not require one scalar utility shared by all brain regions. Evidence for commitment requires prospective generalization and intervention, not only decoding the action after it occurs.

5.1 When one candidate gains effective drive

Consider two ordinary cases. A person may weigh orange juice against milk, or may notice a familiar action beginning and interrupt it. In the first case, partly overlapping populations can represent different expected tastes, values, body needs, and movements. In the second, a well-learned trajectory must become available to additional inhibition or redirection before its usual motor endpoint. Neither case requires a hidden observer, but both require an account of how one distributed candidate becomes more effective than another.

Established physiology already supplies the components. Synaptic input summates at receivers; excitation, inhibition, disinhibition, membrane state, anatomy, and timing change the effect of that input; population activity can prepare action; and descending circuits transform neural recruitment into movement. Phase state can also change an effective motor-cortex interaction under controlled stimulation [34]. The open question is whether a receiver-specific measure that combines these components explains candidate commitment better than ordinary rate, power, anatomy, movement, arousal, and recurrent-state variables.

In plain language, inputs that reach a receiver through effective projections with compatible timing can reinforce one another, while oppositely signed, misaligned, or inhibited inputs can reduce or redirect the net

effect. The result is not muscular energy and does not imply a large net charge in a brain region. It is the signed influence that the presently recruited sources exert on a declared receiver under its current state.

For candidate c , define a measurement-level effective drive

$$D_c(t) = \sum_i w_{ic}(t) I_i(t) \cos(\phi_i(t) - \Phi_c(t)), \quad (5)$$

where $I_i(t)$ is a declared signed current or validated current-sensitive proxy, $\phi_i(t)$ is a declared phase estimate, $\Phi_c(t)$ is a receiver or candidate reference phase, and $w_{ic}(t)$ includes projection efficacy, geometry, orientation, and receiver sensitivity. This quantity is called **coherence-weighted effective drive**. The historical SAN term **electrical valence** is used here only as shorthand for that signed, receiver-specific drive. It is distinct from psychological valence, learned value, salience, membrane voltage, spectral power, chemical valence, and bulk charge.

Let $\bar{D}_c(t)$ be the drive expected under the current tonic, bodily, and task context. The electrical component of a candidate receiver-relative difference is

$$\text{PWD}_{e,c}(t) = D_c(t) - \bar{D}_c(t). \quad (6)$$

Only after the operation is defined does SAN name a sustained, developing departure of this kind a **building phase-wave differential**. The new term does work that generic phase coding and traveling-wave descriptions do not do by themselves: it binds a sender-receiver relation, an expected tonic context, a signed effective consequence, and a candidate route. It remains a hypothesis. If Equations (5)-(6) add no held-out or intervention-level value beyond the declared comparators, electrical valence and the building-PWD component are removed while the ordinary route-selection model remains.

5.2 Bounded preparation-action-return anchor

Bat vocalization illustrates the full ordering without proving the proposed variable. Pre-vocal beta and high-gamma features in frontal and striatal recordings differed with the impending call class and supported call classification [35]. This was not one universal beta-gamma anticorrelation: the reported effects depended on region, depth, epoch, and call class. A later study found predominantly frontal-to-auditory directed interaction before vocalization and an auditory-to-frontal reversal after echolocation [36]. Together, the studies measure preparation-specific oscillatory structure, prospective action information, and returned sensory flow. They do not show that one frequency band commands another or that an LFP directly contracts a vocal fold.

The bounded sequence is:

```
candidate state
-> receiver-compatible effective drive
-> changed route accessibility
-> population commitment
-> descending motor realization
-> movement or vocal output
-> sensory and bodily return
-> later criterion or policy update
```

This is the mechanism bridge carried by the Choice paper. A separate coherence-and-motor analysis may test the electrical term in greater detail; the present paper requires only that every arrow retain its own measurement, intervention, and failure condition.

6. Sight to Action and Back

A visual choice makes the loop concrete. Current visual evidence is interpreted with a body / world estimate, task context, recurrent expectation, and the consequences of preceding actions. That state changes action-related population dynamics and effective connectivity. After commitment, a motor command and efference copy support a prediction of the next sensory state. Movement changes the scene and body. Reafference then updates the body / world estimate, policy, and potentially the criteria governing the next selection.

After action a_t^* , the update can be written as

$$(\mathbf{w}, \boldsymbol{\theta}, \mathbf{c})_{t+1} = U((\mathbf{w}, \boldsymbol{\theta}, \mathbf{c})_t, \hat{\mathbf{y}}_t, \mathbf{y}_t, \mathbf{r}_t), \quad (7)$$

where $\hat{\mathbf{y}}_t$ is the predicted consequence, $\mathbf{y}_t$ the observed consequence, and $\mathbf{r}_t$ a declared reward, error, or other feedback variable. The update may occur through multiple plasticity mechanisms and timescales. It is not identified with one dopamine signal or one universal learning rule. Dendritic activity can causally affect perception in bounded preparations, and behavioral-timescale plasticity can alter later responses rapidly [10,11], but neither result establishes the full choice ladder.

The detailed biological-vision and reafference mechanism belongs to the published embodied neural-rendering paper [19]. The present paper owns the question of how that conditioned state changes route, competition, commitment, and later policy. The two mechanisms can fail independently.

Neural recruitment reaches muscle through additional transformations rather than by exporting an oscillatory band as mechanical energy. Depending on the action, cortical and subcortical activity recruits brainstem or spinal motor circuits, motor neurons, neuromuscular transmission, calcium-dependent muscle activation, and coordinated agonist, antagonist, stabilizing, respiratory, or laryngeal mechanics. Comparative tracing confirms that skilled vocalization uses distributed cortical and premotor routes to laryngeal motoneurons [37]. The physiological energy for contraction is supplied locally through cellular metabolism; coincident neural activity helps determine whether the motor pathway crosses its activation conditions.

7. Optional Receiver-Relative Timing Residual

Rhythms can organize excitability and communication windows, but generic phase or synchrony is not evidence for this paper's stronger residual. The incremental comparison is

$$\Delta_{\text{timing}} = \text{Score}(M_{\text{base}} + \Phi_{\text{rel}}) - \text{Score}(M_{\text{base}}), \quad (8)$$

where M_{base} includes anatomy, rate, power, movement, arousal, task variables, waveform controls, common-input controls, and a capacity-matched recurrent state. Φ_{rel} contains preregistered receiver-relative phase, duration, magnitude, and spatial relations. The extension earns support only if it generalizes to held-out biological units and survives a selective perturbation and rescue. If it fails, the ordinary criteria-route-commitment-feedback ladder remains testable.

8. Evidence and Model Alternatives

8.1 Distributed cortico-subcortical implementation

The route model cannot be made cortex-only. In dorsal striatum, brief cell-type-specific stimulation can bias choice distributions in a reward-history-dependent manner [21]. Indirect-pathway striatal neurons can also change exploration through collicular competition [30]. These results support circuit-specific selection effects, but they do not justify a simple direct-equals-go and indirect-equals-stop rule. In a different preparation, closed-loop stimulation of direct and indirect pathways changed movement velocity without changing action selection [31].

The superior colliculus supplies another bounded route. GABAergic collicular neurons contributed to spatial choice through functionally long-range rather than purely local inhibition [14]. That result is especially important for the historical “train switch” analogy: inhibition can redistribute downstream accessibility, but the biological route, target, and spatial scale must be measured rather than inferred from the metaphor.

Thalamic results split at least three functions that a one-node chooser would collapse. Mediodorsal thalamus supported switching between learned cueing contexts and stabilized context-relevant prefrontal representations [22]. Mediodorsal-thalamus lesions impaired rapid reward-guided updating in changing environments [29]. Motor-thalamus populations separately tracked action initiation, selection, and vigor in a two-alternative task [24]. These are preparation-bounded functions, not evidence that the thalamus is the chooser.

Cerebellar recording and perturbation during evidence accumulation found choice- and evidence-related Purkinje-cell activity and reduced performance after inactivation [23]. This supports cerebellar participation in a distributed choice loop. It does not identify a universal decision variable or show that cerebellar activity alone determines commitment.

8.2 Comparator matrix

Comparator	What it already explains	Discriminator required here
evidence accumulation	noisy integration toward a boundary	route variables must add held-out or causal value
recurrent attractors	competition, persistence, categorical commitment	route change must differ from ordinary state or attractor depth
affordance competition	simultaneous action specification and selection	scale ladder must add mechanism rather than terminology
context-dependent dynamics	flexible trajectories in recurrent state space	identified route or timing variables must improve intervention prediction
basal-ganglia gating	action-value biases, suppression, exploration, and movement control through subcortical loops [21,30,31]	a route claim must specify pathway, target, task epoch, and whether it changes selection, vigor, or execution
superior-colliculus competition	spatial target selection and long-range inhibitory competition [14]	generalization beyond the measured spatial task requires a new intervention
thalamocortical regulation	context switching, representational maintenance, reward-guided updating, initiation, and vigor [22,24,29]	separate cortical maintenance, policy updating, commitment, and motor execution
cerebellar contribution	evidence-related activity, sensorimotor prediction, and decision-task performance [23]	distinguish evidence processing, execution correction, and commitment
active inference	action and perception as inference under expected sensory consequences [26]	show that a route variable adds intervention-level prediction beyond a fitted generative-policy model

Comparator	What it already explains	Discriminator required here
dynamic causal modeling	context-dependent effective coupling under an explicit causal model [25]	the proposed route must exceed a declared DCM or equivalent estimator, not merely rename effective connectivity
network control	structural constraints on possible state transitions [27]	report model and null sensitivity; structure-only controllability is not trial-specific causal routing [28]
critical causation	information-dependent modification of later criteria	retain Tse’s priority and test the later multiscale join
readiness-potential models	preparation before reported intention	distinguish commitment, report, and averaged preparation
sense-of-agency models	prediction and attribution of control	separate causal control from subjective ownership

An ordinary comparator that performs equally well is not a nuisance result. It narrows or removes the proposed addition.

8.3 Positive, null, and contradictory evidence

Positive evidence is strongest for individual links, not for the whole ladder. Dendritic perturbation, cell-class-specific inhibition, striatal choice bias, collicular competition, thalamic switching, cerebellar task contribution, and reafferent updating each make a local arrow biologically plausible [10-17,20-24,29,30]. None of those findings establishes the joined criteria-to-route-to-commitment-to-consequence mechanism.

Several findings actively constrain the proposal. Direct and indirect striatal pathways can modulate velocity without changing action selection [31]. Superior-colliculus inhibition can act through long-range interactions rather than local suppression [14]. Readiness-potential averages can emerge from a stochastic accumulator without fixing one hidden decision time [8]. Network controllability results depend on the dynamical model and null ensemble, and single-node interpretations have been challenged [27,28]. A DCM, recurrent state-space model, active-inference model, or ordinary attractor that predicts the perturbation equally well removes the need for a new route construct.

8.4 Nested acceptance and removal rules

Let held-out loss be $\mathcal{L}$. The incremental route score is

$$\Delta_R = \mathcal{L}(M_{\text{base}}) - \mathcal{L}(M_{\text{base}} + R), \quad (9)$$

where M_{base} includes the declared task, movement, arousal, history, intervention, and ordinary recurrent or effective-connectivity variables. A route is accepted only when it is prospective, improves held-out performance beyond a frozen threshold, responds to the declared intervention, and predicts a downstream redistribution rather than only the eventual label.

The timing residual is independently removable:

$$\Delta_\Phi = \mathcal{L}(M_{\text{base}} + R) - \mathcal{L}(M_{\text{base}} + R + \Phi_{\text{rel}}). \quad (10)$$

If Δ_Φ fails the frozen held-out gate, or if its apparent value depends on rate, power, movement, arousal, waveform, common input, or post-outcome information, Φ_{rel} is removed from the scientific model. Its historical SAN label may remain in the genealogy but cannot carry a biological claim.

9. Experimental Program

9.1 Criteria-route-commitment recording

Use a task with explicit alternatives, context reversals, and stable sensory evidence. Combine compartment-sensitive or dendritic measurement, identified interneuron recording, population recording, movement tracking, pupil / arousal measurement, and behavior. Estimate structural, functional, and effective networks separately. Test temporal precedence and whether trial-specific route estimates mediate context-to-commitment changes.

9.2 Cell-class-specific intervention

Perturb a declared interneuron class at a declared target and time window. Predict the downstream population and action-distribution change before unblinding. Include neighboring-time, neighboring-class, sham, movement, arousal, sensation, and general-performance controls. Global impairment does not establish route selection.

9.3 Dendritic criterion intervention

Manipulate a dendritic event or learned criterion while holding the immediate stimulus and gross network as constant as feasible. Test the predicted order: dendritic state, effective route, population commitment, and behavior. Separate the result from nonspecific excitability.

9.4 Consequence reversal

Reverse action consequences without changing sensory discrimination. Test whether predictions, route estimates, candidate trajectories, behavior, and later criteria change in the preregistered order. Separate reward magnitude, surprise, error, punishment, movement correction, and explicit rule learning.

9.5 Timing residual and rescue

Compare the locked baseline with the same model plus receiver-relative timing. Where technically feasible, manipulate timing while approximately preserving rate, mean power, stimulus, sensation, and movement. Require target engagement, endpoint specificity, and a timing-specific rescue.

9.6 Cross-task and cross-scale generalization

Train route and commitment models in one context and test them on another. A general operator should survive superficial stimulus changes while retaining its declared circuit and scale. Failure can narrow the proposal to one preparation rather than falsifying all biological choice.

9.7 Identifiability matrix

The route variable is not accepted merely because it predicts the eventual action. Each experiment must declare the following chain before unblinding:

Link	Required observation	Required intervention	Failure interpretation
criterion to cellular transition	criterion-dependent dendritic or membrane event before population commitment	alter the criterion-bearing input while matching immediate stimulus and gross excitability	no identified cellular criterion term
inhibition to effective route	cell-class, compartment, and time-specific change in downstream accessibility	perturb the declared inhibitory class in the declared window	global gain or impairment is not route selection
effective route to commitment	route estimate prospectively predicts which candidate population stabilizes	disrupt the route after candidate representation but before commitment	retrospective decoding cannot carry the arrow
commitment to action	committed state controls a declared motor output under matched alternatives	perturb the committed population or its output pathway	correlation without behavioral control
consequence to later criterion	reafferent mismatch or outcome changes the next declared criterion or policy	reverse or clamp the consequence while holding discrimination constant	generic reward, surprise, or relearning remains sufficient

The full ladder is supported only if every required arrow survives its own intervention and the mediation order is consistent with the declared temporal sequence. A successful cell-class perturbation may validate the inhibitory link while leaving population commitment, agency, and later updating open.

9.8 Deterministic synthetic identifiability fixture

The executable fixture in `model/run_draft3_identifiability.py` uses 30 deterministic replicates of 1,800 synthetic trials for each of five scenarios. A regularized logistic estimator compares a frozen baseline against a baseline-plus-route model on a held-out split. Acceptance additionally requires prospective provenance and a minimum intervention effect on the route variable. A nested timing model is evaluated separately. No participant, animal, imaging, or electrophysiological data are used.

Scenario	Mean held-out route gain	Mean intervention SMD	Acceptance	Interpretation
true prospective route	0.0352	0.6205	30/30	fixture recovers a declared route effect
route null	-0.0004	0.6415	0/30	intervention shift alone is insufficient
global-gain confound	-0.0002	0.6659	0/30	matching intervention and nuisance terms blocks false routing
post-outcome leakage	0.5525	0.0136	0/30	high prediction is rejected because provenance is invalid
timing null	route gain 0.0374; timing gain -0.0003	0.6113	timing 0/30	ordinary route retained; timing residual removed

All eleven software gates pass. The deterministic payload SHA-256 is 159B3277E8067B14B0783C3D764AA0A6ABC1153A9138CE84F8DF0F71E80A9576. The fixture establishes that the declared acceptance logic can reject four important failure modes under known generated conditions. It does not show that a corresponding biological variable exists, is measurable, or is unique.

9.9 Frozen IBL Brain Wide Map route-screen plan

This version carries a machine-readable analysis plan at `protocol/ibl-brainwidemap-route-screen-preregistration-v1.json`. It targets the International Brain Laboratory’s public Brainwidemap release, which contains 459 sessions, 699 probe insertions, 621,733 sorted units, 75,708 quality-passed units, 139 mice, stimulus and choice variables, response times, wheel traces, and pose estimates [32,33]. No IBL data were downloaded, opened, fitted, or inspected while this plan was written.

The screen is deliberately weaker than the causal experiments in Sections 9.1-9.5. Eligible sessions must contain at least two simultaneous probe insertions in distinct areas, quality-passed spike data, task variables, wheel data, and precommitment pose covariates. Trials are nested within sessions and sessions within animals. Animals, rather than trials, are assigned by a deterministic identifier hash to 60% training, 20% calibration, and 20% untouched test partitions. Source-target area families, lags, regularization, and all decision thresholds must be selected using training and calibration animals and then locked before one evaluation on the animal-held-out test set.

The baseline includes stimulus side and contrast, block prior, prior choice and outcome, reaction-time-matched wheel and pose variables, target history, session and animal structure, population rate, spectral power where available, and a capacity-matched recurrent or reduced-rank neural-state model. A candidate route term is restricted to lagged source-to-target information available before the first wheel movement. It must improve both held-out choice log loss and a separate downstream-redistribution endpoint beyond that baseline. Condition- matched time shifts, route-direction reversal, area-label permutation, global gain, movement / arousal sensitivity, and post-outcome leakage are mandatory nulls. The optional receiver-relative timing residual in Section 7 is excluded from this first biological screen; it can be tested only after an ordinary route term survives.

Acceptance requires a positive animal-level 95% bootstrap confidence interval for both preregistered gains on the untouched test set, consistency under the locked nuisance analyses, and failure of the declared nulls. The plan does not permit selecting a favorable area pair, lag, endpoint, or threshold after viewing test results. Even a positive screen would establish only incremental prospective association in this observational dataset. It would not identify a cell class, prove inhibitory path diversion, establish the full causal ladder, demonstrate consciousness, or show moral responsibility. A negative screen would reject the frozen route estimator on this task and dataset while leaving more preparation-specific causal tests open.

10. Claim-Level Failure Conditions

Claim ID	Candidate claim	Failure or narrowing condition
CHC-04	structural and effective routes are experimentally separable	route estimates are unstable or merely restate the action label
CHC-05	identified inhibition changes a specific downstream route	effect reduces to global gain, arousal, sensation, or motor impairment
CHC-08	multiple candidate actions coexist before commitment	task does not establish alternatives or a serial model predicts equally well
CHC-09	organism choice is a bounded transition to controlled action	commitment is only retrospective decoding and does not control behavior
CHC-10	consequences change later criteria or policy	only generic reward / error predicts updating
CHC-13	the full ladder adds explanatory value	disconnected or ordinary recurrent models generalize equally well
CHC-14	receiver-relative timing adds information	matched baseline wins or added value is analysis-dependent

Claim ID	Candidate claim	Failure or narrowing condition
CHC-15	timing intervention changes route specifically	behavior changes only through rate, power, sensation, or movement
CHC-16	the inhibitory route term is identifiable	cell-class perturbation changes only global excitability or the eventual action label without a prospective downstream redistribution
CHC-21	coherence-weighted effective drive adds a receiver-specific commitment variable	anatomy, rate, power, movement, arousal, common input, geometry-free coherence, or a capacity-matched recurrent model performs equally well
CHC-22	a candidate route is connected through motor realization to returned consequence	the candidate signal does not prospectively mediate motor recruitment, or the returned consequence does not alter the declared later state

10.1 Internal adversarial disposition

This pass is an internal red-team review. External neuroscience, methods, philosophy, and terminology review remains valuable, but it is not represented as completed and is not a local completeness gate for this theory-and-protocol preprint.

Challenge	Disposition retained in this version	Remaining burden
The model merely renames effective connectivity	Retain <i>effective route</i> only as an estimator-bound construct that must add prospective and perturbational value beyond ordinary effective-connectivity models	Remove it if Equations (9)-(10) do not survive biological data
Inhibitory path diversion is treated too literally	State changes in gain, excitability windows, competition, and downstream trajectory are the measurable claim; no signal is assumed to travel through a unique railway-like path	Name cell class, target, timing, and behavioral endpoint in every intervention
Choice, action selection, commitment, and agency are conflated	Keep the four-scale ladder and treat commitment as control of a task-declared action under maintained alternatives	Predeclare the action set, commitment window, and causal-control criterion
Phase variables immunize the theory from ordinary nulls	Keep the timing residual optional and removable	Rate, power, waveform, movement, arousal, common-input, and capacity-matched recurrent baselines remain mandatory
Historical computer-network language is mistaken for biology	Confine TCP, UDP, and Alternating Transmission Protocol to genealogy and Appendix A as functional analogies	No Internet packet, handshake, retransmission, or reliability semantics may be attributed to neural tissue without direct evidence

11. Genealogy and Source Boundaries

The authorial genealogy contains nineteen bounded atoms from 2011 through 2025. Owner dates, public fixation dates, archive dates, Git commits, and later modification dates remain separate. The chronology supports the development of Micah Blumberg’s synthesis; it does not validate its biology.

The relevant public sequence includes a 2017 Neural Lace / NerveGear discussion of criteria, dendritic charge, routing, and action. In the same podcast program, Micah asked whether brain communication

was functionally more like connection-and-feedback transmission or connectionless broadcast, using TCP and UDP as computer-network analogies. He proposed **Alternating Transmission Protocol (ATP)** for a hybrid that could switch between those functions and chose the acronym deliberately because adenosine triphosphate is the cell's energy currency. This is a protocol-search ancestor, not evidence that neurons implement Internet protocols and not the completed PWD or NAPOT mechanism. The sequence continues through 2018 Medium / SVGN bridges on mechanical biological agency and dendritic criteria, a 2021 synaptic-unreliability bridge, and a 2022 public circuit-scale formulation joining inhibitory path bifurcation to choice. Buzsaki and Tse remain named, bounded donors. The later synthesis does not retroactively absorb their atoms.

Historical Self-Aware Networks terms are preserved in Appendix A and the governed source packet. They are not required to understand the scientific body.

12. Scope

This paper does not claim:

- one universal choice neuron, interneuron class, cortical region, or route;
- that a decoded future action fixes an unconscious decision time;
- that randomness creates authorship;
- that determinism eliminates useful organism-level regulation;
- that every choice is conscious;
- that choice proves phenomenal consciousness;
- that information acts without a physical implementation;
- that a neural mechanism derives moral or legal responsibility;
- that prior authors endorse this joined synthesis;
- that electrical valence is psychological valence, chemical valence, bulk charge, spectral power, or energy delivered to muscle; or
- that one beta, gamma, or other frequency band directly contracts a muscle or universally commands another band.

13. Conclusion

Choice can be studied without a central chooser by tracking a causal sequence across declared scales and circuits. Learned criteria may change cellular transitions; state-dependent cortical and subcortical conditions may alter effective influence; distributed populations may move toward commitment; a bounded organism controls an action; and the action's consequences can update later behavior. This is a falsifiable model, not a metaphysical exemption. The deterministic fixture shows that the acceptance logic can recover a generated route while rejecting route-null, global-gain, leakage, and timing-null fixtures. It also shows why the empirical burden is harder: basal ganglia, superior colliculus, thalamus, cerebellum, recurrent dynamics, active inference, effective connectivity, and network control already explain substantial parts of the sequence. This final-preprint successor exposes that logic to a frozen public-data screen without mistaking association for intervention. The joined ladder survives only if each arrow ultimately adds prospective and, where causality is claimed, intervention-based value beyond those alternatives.

Appendix A. Historical SAN Concordance

Historical label	Main-text operational counterpart	Status
Alternating Transmission Protocol (ATP)	historical question about switching between feedback-dependent and broadcast-like neural communication	2017 genealogy only; TCP / UDP are functional analogies, and the acronym intentionally also invokes cellular ATP
Choice as a Dynamic Neuropath	time-bounded effective route and population transition	genealogy only; retire if it adds ambiguity
Choice Refinement	consequence-dependent criterion or policy update	testable synthesis
Gamma Consideration Sandwich	sensory / context / action timing interaction	cross-paper hypothesis, not required here
electrical valence	signed, receiver-specific coherence-weighted effective drive	optional and removable mechanism bridge; not psychological valence or bulk charge
phase-wave differential	receiver-relative transient timing / magnitude / current / context feature	optional residual; a building electrical component is defined only after the plain-language operation
neural rendering	recurrent sensory-state construction and redistribution	owned in detail by published SAN papers
NAPOT	multiscale convergence, integration, and redistribution	broader SAN program; not an explanatory primitive here

Appendix B. Source Grades

A denotes public timestamp or immutable Git evidence; A- adds a bounded identity or archive caveat; B denotes an owner-dated source preserved later in public Git; and C denotes retrospective or AI-assisted material unable to carry an early date by itself. The full nineteen-atom matrix and immutable event ledger remain in CHOICE-ATOM-SOURCE-MATRIX.md.

Appendix C. Source-Locator Audit

The Buzsaki source is bounded to Chapter 11, Figure 11.5 and its caption in *Rhythms of the Brain*. The recovered Kindle screenshot is at 65% progress and does not display a stable print-page number. The caption states that neuronal activity in a multidimensional gridspace is shaped by synaptic strengths and momentary inhibition, and that inhibition at a junction can divert the activity trajectory. The manuscript therefore cites the figure and chapter rather than inventing a print-page locator. The screenshots are provenance evidence and are not publication-cleared figures.

The Tse locator is now closed for every atom retained in the scientific body. Sections 3.9-3.10 (pp. 25-26) supply the set / reset, variable-input, and later criterion-satisfaction sequence, including the distinction between a present physical realization and alteration of future criteria. Sections 4.1-4.4 (pp. 31-34) supply the neuronal physical / informational-criterion and coincidence-sensitive firing context. Micah's 2014 note remains a paraphrase, and the manuscript does not replace Tse's more precise temporal distinction with that later wording.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg's original research, developed through a

continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT's public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper's arguments, interpretations, and errors. # References

1. Buzsaki, G. *Rhythms of the Brain*. Oxford University Press (2006). DOI 10.1093/acprof:oso/9780195301069.001.0001.
2. Tse, P. U. *The Neural Basis of Free Will: Criterial Causation*. MIT Press (2013). DOI 10.7551/mitpress/9780262019101.001.0001.
3. Cisek, P. & Kalaska, J. F. Neural correlates of reaching decisions in dorsal premotor cortex. *Neuron* 45, 801-814 (2005). DOI 10.1016/j.neuron.2005.01.027.
4. Cisek, P. Cortical mechanisms of action selection: the affordance competition hypothesis. *Philosophical Transactions B* 362, 1585-1599 (2007). DOI 10.1098/rstb.2007.2054.
5. Wang, X.-J. Probabilistic decision making by slow reverberation in cortical circuits. *Neuron* 36, 955-968 (2002). DOI 10.1016/S0896-6273(02)01092-9.
6. Gold, J. I. & Shadlen, M. N. The neural basis of decision making. *Annual Review of Neuroscience* 30, 535-574 (2007). DOI 10.1146/annurev.neuro.29.051605.113038.
7. Mante, V. et al. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature* 503, 78-84 (2013). DOI 10.1038/nature12742.
8. Schurger, A. et al. An accumulator model for spontaneous neural activity prior to self-initiated movement. *PNAS* 109, E2904-E2913 (2012). DOI 10.1073/pnas.1210467109.
9. Haggard, P. Sense of agency in the human brain. *Nature Reviews Neuroscience* 18, 196-207 (2017). DOI 10.1038/nrn.2017.14.
10. Takahashi, N. et al. Active cortical dendrites modulate perception. *Science* 354, 1587-1590 (2016). DOI 10.1126/science.aah6066.
11. Bittner, K. C. et al. Behavioral time scale synaptic plasticity underlies CA1 place fields. *Science* 357, 1033-1036 (2017). DOI 10.1126/science.aan3846.
12. Pi, H.-J. et al. Cortical interneurons that specialize in disinhibitory control. *Nature* 503, 521-524 (2013). DOI 10.1038/nature12676.
13. Lee, S. et al. A disinhibitory circuit mediates motor integration in the somatosensory cortex. *Nature Neuroscience* 16, 1662-1670 (2013). DOI 10.1038/nn.3544.
14. Essig, J. et al. GABAergic activity in the superior colliculus contributes to choice behavior. *Communications Biology* 4 (2021). DOI 10.1038/s42003-021-02248-1.
15. Roach, J. P. et al. Choice-selective inhibition stabilizes decision circuits. *Nature Communications* 14 (2023). DOI 10.1038/s41467-023-35822-8.
16. Kuan, A. T. et al. Opponent excitatory and inhibitory wiring motifs in mouse posterior parietal cortex. *Nature* (2024). DOI 10.1038/s41586-024-07088-7.
17. Maristany de las Casas, M. et al. Dendritic inhibition shapes flexible rule relearning. *Science* (2026). DOI 10.1126/science.adx4358.
18. Blumberg, M. *Self Aware Networks*, Version 2.1 (2026). DOI 10.5281/zenodo.21329627.
19. Blumberg, M. *Embodied Neural Rendering in Self-Aware Networks*, Version 1.1 (2026). DOI 10.5281/zenodo.21331180.
20. Lovett-Barron, M. et al. Regulation of neuronal input transformations by tunable dendritic inhibition. *Nature Neuroscience* 15, 423-430 (2012). DOI 10.1038/nn.3024.
21. Tai, L.-H. et al. Transient stimulation of distinct subpopulations of striatal neurons mimics changes in

- action value. *Nature Neuroscience* 15, 1281-1289 (2012). DOI 10.1038/nn.3188.
22. Rikhye, R. V., Gilra, A. & Halassa, M. M. Thalamic regulation of switching between cortical representations enables cognitive flexibility. *Nature Neuroscience* 21, 1753-1763 (2018). DOI 10.1038/s41593-018-0269-z.
 23. Devereett, B. et al. Cerebellar involvement in an evidence-accumulation decision-making task. *eLife* 7, e36781 (2018). DOI 10.7554/eLife.36781.
 24. Gaidica, M. et al. Distinct populations of motor thalamic neurons encode action initiation, action selection, and movement vigor. *Journal of Neuroscience* 38, 6563-6573 (2018). DOI 10.1523/JNEUROSCI.0463-18.2018.
 25. Friston, K. J., Harrison, L. & Penny, W. Dynamic causal modelling. *NeuroImage* 19, 1273-1302 (2003). DOI 10.1016/S1053-8119(03)00202-7.
 26. Friston, K. J. et al. Action and behavior: a free-energy formulation. *Biological Cybernetics* 102, 227-260 (2010). DOI 10.1007/s00422-010-0364-z.
 27. Gu, S. et al. Controllability of structural brain networks. *Nature Communications* 6, 8414 (2015). DOI 10.1038/ncomms9414.
 28. Tu, C. et al. Warnings and caveats in brain controllability. *NeuroImage* 176, 83-91 (2018). DOI 10.1016/j.neuroimage.2018.04.010.
 29. Chakraborty, S. et al. Critical role for the mediodorsal thalamus in permitting rapid reward-guided updating in stochastic reward environments. *eLife* 5, e13588 (2016). DOI 10.7554/eLife.13588.
 30. Lee, J. & Sabatini, B. L. Striatal indirect pathway mediates exploration via collicular competition. *Nature* 599, 645-649 (2021). DOI 10.1038/s41586-021-04055-4.
 31. Tecuapetla, F. et al. Opponent and bidirectional control of movement velocity in the basal ganglia. *Nature* 533, 532-536 (2016). DOI 10.1038/nature17639.
 32. International Brain Laboratory et al. A brain-wide map of neural activity during complex behaviour. *Nature* 645, 177-191 (2025). DOI 10.1038/s41586-025-09235-0.
 33. International Brain Laboratory. [2025 Brain Wide Map data release documentation](#), release tag Brainwidemap (accessed 20 July 2026).
 34. Stefanou, M.-I. et al. Phase synchronicity of mu-rhythm determines efficacy of interhemispheric communication between human motor cortices. *Journal of Neuroscience* 38, 10525-10534 (2018). DOI 10.1523/JNEUROSCI.1470-18.2018.
 35. Weineck, K., Garcia-Rosales, F. & Hechavarria, J. C. Neural oscillations in the fronto-striatal network predict vocal output in bats. *PLOS Biology* 18, e3000658 (2020). DOI 10.1371/journal.pbio.3000658.
 36. Garcia-Rosales, F. et al. Echolocation-related reversal of information flow in a cortical vocalization network. *Nature Communications* 13, 3642 (2022). DOI 10.1038/s41467-022-31230-6.
 37. Cerkevich, C. M., Rathelot, J.-A. & Strick, P. L. Cortical basis for skilled vocalization. *Proceedings of the National Academy of Sciences* 119, e2122345119 (2022). DOI 10.1073/pnas.2122345119.